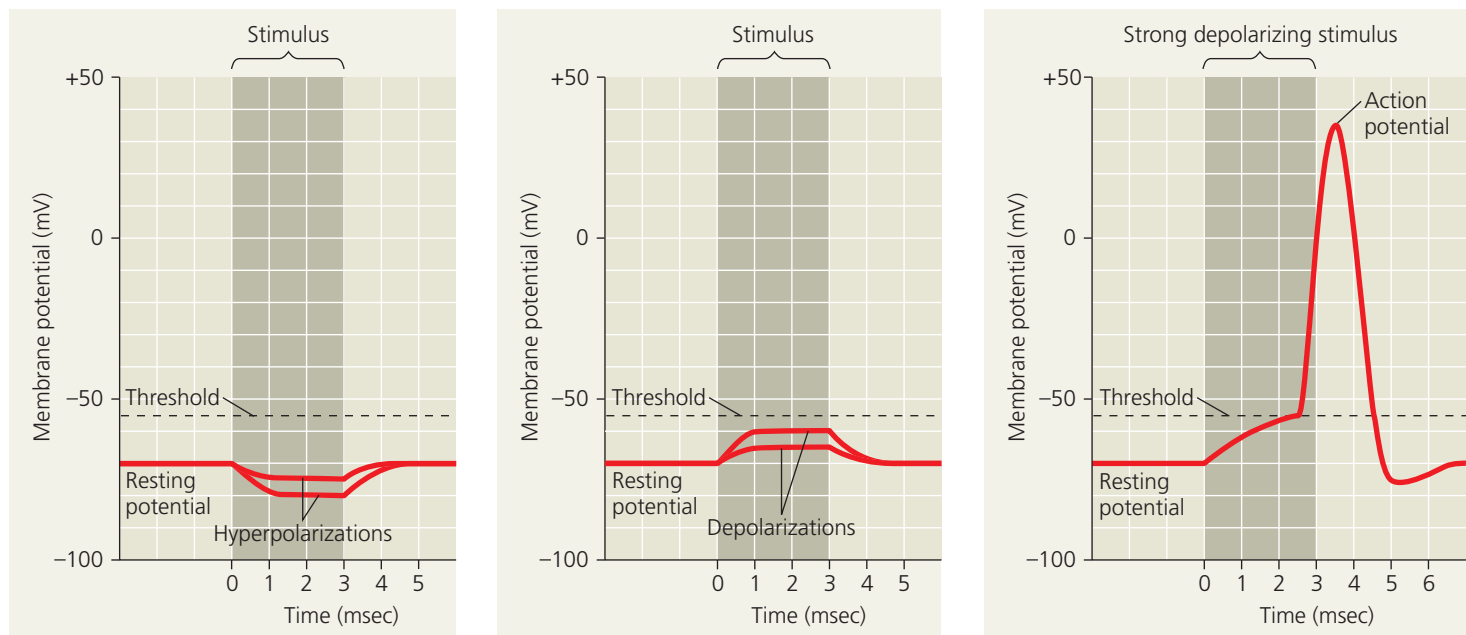


▼ **Figure 48.11** Graded potentials and an action potential in a neuron.



(a) Graded hyperpolarizations produced by two stimuli that increase membrane permeability to K^+ . The larger stimulus produces a larger hyperpolarization.

(b) Graded depolarizations produced by two stimuli that increase membrane permeability to Na^+ . The larger stimulus produces a larger depolarization.

(c) Action potential triggered by a depolarization that reaches the threshold.

DRAW IT Redraw the graph in (c), extending the y-axis. Then label the positions of E_K and E_{Na^+} .

magnitude of the membrane potential is a **depolarization**. In neurons, depolarization often involves gated sodium channels. If a stimulus causes gated sodium channels to open, the membrane's permeability to Na^+ increases. Na^+ diffuses into the cell along its concentration gradient, causing a depolarization as the membrane potential shifts toward E_{Na^+} (+62 mV at 37°C).

Graded Potentials and Action Potentials

Sometimes, the response to hyperpolarization or depolarization is simply a shift in the membrane potential. This shift, called a **graded potential**, has a magnitude that varies with the strength of the stimulus: A larger stimulus causes a greater change in the membrane potential (see Figure 48.11a and 48.11b). Graded potentials induce a small electrical current that dissipates as it flows along the membrane. Graded potentials thus decay with time and with distance from their source.

If a depolarization shifts the membrane potential sufficiently, the result is a massive change in membrane voltage called an **action potential**. Unlike graded potentials, action potentials have a constant magnitude and can regenerate in adjacent regions of the membrane. Action potentials can therefore spread along axons, making them well suited for transmitting a signal over long distances.

Action potentials arise because some of the ion channels in neurons are voltage-gated (see Figure 48.10). If a depolarization increases the membrane potential to a level

called **threshold**, the voltage-gated sodium channels open. The resulting flow of Na^+ into the neuron results in further depolarization. Because the sodium channels are voltage gated, the increased depolarization causes more sodium channels to open, leading to an even greater flow of current. The result is a process of positive feedback that triggers a very rapid opening of many voltage-gated sodium channels and the marked temporary change in membrane potential that defines an action potential (**Figure 48.11c**).

The positive-feedback loop of channel opening and depolarization triggers an action potential whenever the membrane potential reaches threshold, about -55 mV for many mammals. Once initiated, the action potential has a magnitude that is independent of the strength of the triggering stimulus. Because action potentials either occur fully or do not occur at all, they represent an *all-or-none* response to stimuli.

Generation of Action Potentials: A Closer Look

The characteristic shape of the action potential graph in Figure 48.11c reflects changes in membrane potential resulting from ion movement through voltage-gated sodium and potassium channels. Depolarization opens both types of channels, but they respond independently and sequentially. Sodium channels open first, initiating the action potential. As the action potential proceeds, sodium channels remain open